Indoor 3D Visible Light Positioning Based on Multiple Cameras: Algorithm Design and Error Analysis

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***Abstract*— In this paper, a novel multiple-camera-based visible light positioning (MC-VLP) algorithm is proposed for indoor three-dimensional (3D) positioning. The basic idea of MC-VLP is to utilize multiple cameras to simultaneously capture the beacon signal transmitted by a light-emitting diode (LED) mounted on the mobile terminal. Since the pinhole camera can achieve high imaging accuracy and the fish-eye camera has a wide field of view (FOV), both of them can be adopted as the receiver to improve the positioning accuracy and coverage area. Meanwhile, the positioning errors along X-axis, Y-axis, and Z-axis are derived to evaluate the performance of the MC-VLP algorithm. Based on the derived expression of positioning error, we optimize the camera layout and tilt angle to minimize the positioning error in the coverage area. Simulation results show that higher positioning accuracy can be achieved via optimizing the camera layout and tilt angle. Experimental results show that the mean 3D positioning accuracy is 1.7 mm with 3 pinhole cameras at height 2.5 m.**

***Index Terms*— Visible light positioning (VLP), multiple cameras, three-dimensional (3D) positioning, pinhole camera, fish-eye camera, positioning error.**

I. Introduction

HIGH-ACCURACY indoor positioning technology has received extensive research attention recently due to the growing demand in indoor location-based services (ILBS) such as indoor navigation, virtual reality, unmanned vehicle, and robot control. Until now, plenty of indoor positioning technologies have been proposed, such as radio frequency identification (RFID) [1], ultra-wideband (UWB) [2], Bluetooth [3], and WiFi [4]. However, the positioning accuracies of these technologies are mostly in the meter- or decimeter-level, which cannot meet the indoor positioning requirements.

Visible light positioning (VLP) technologies utilize the visible light signals transmitted by light emitting diodes (LEDs) to determine the position of mobile terminal [5], [6], [7]. Since the widely deployed LEDs are adopted as beacons, VLP has relatively low deployment cost. Besides, due to strong directionality of visible light, VLP can achieve higher positioning accuracy than traditional positioning approaches. In the reported VLP systems, photodiode (PD) or camera is adopted as the receiver. Positioning algorithms using PD include triangulation, fingerprint identification [8], and proximity detection [9]. The triangulation method utilizes the distance or angle between the positioning point and the beacons with known positions to calculate the coordinate of the terminal by time of arrival (TOA) [10], time difference of arrival (TDOA) [11], angle of arrival (AOA) [12], or received signal strength (RSS) [13]. However, PD-based VLP is easily affected by ambient light and the direction of light beam, which may significantly limit the mobility of users and the robustness of VLP system.

Meanwhile, a number of camera-based VLP technologies have been proposed, which determine the receiver position based on the geometric relations between cameras and LEDs. A single camera and multiple LEDs can be adopted for VLP. For instance, three-dimensional (3D) VLP is realized in [14] according to the angle difference of arrival (ADOA). A perspective-four-line algorithm is proposed in [15] to estimate the receiver position using a single parallelogram LED array and a camera. Work [16] verified a 3D positioning accuracy of 5 cm via simulations under 3-meter height. Two positioning approaches are adopted in [17] for two scenarios where the LED positions are known and unknown. In [18], the receiver position and orientation are calculated using singular value decomposition (SVD). VLP systems can also adopt single camera, single LED, and an inertial measurement unit (IMU). Work [19] utilized the geometric features of a circular LED's projection on image sensor to estimate the receiver position. A loosely-coupled VLP-inertial fusion method is proposed in [20] using extended Kalman filter (EKF). Work [21] compared sensor fusion approaches based on EKF, particle filter, and a combined filter. Besides, to simultaneously exploit visual and strength information of visible lights, a camera assisted received signal strength (CA-RSS) algorithm and an enhanced camera assisted received signal strength ratio (eCA-RSSR) algorithm are proposed in [22] and [23], respectively. Moreover, sever-based VLP systems [24] can adopt multiple cameras and a single LED, works [25] and [26]

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TABLE I

EXISTING EXPERIMENTAL RESULTS OF CAMERA-BASED VLP ALGORITHMS

Work	Server-side devices	User-side device	Height	Mean accuracy
[27]	4 LEDs	1 fish-eye camera	3.1 m	17 cm
[28]	6 LEDs	1 fish-eye camera	2.1 m	10 cm
[14]	4 LEDs	1 pinhole camera	2.5 m	3.2 cm
[15]	4 LEDs	1 pinhole camera	1.75 m	2.73 cm
[25]	2 pinhole cameras	1 LED	1.6 m	7.6 mm
[26]	2 pinhole cameras	1 LED	2 m	2.9 mm
Ours	2 pinhole cameras	1 LED	2.5 m	2.6 mm
Ours	3 pinhole cameras	1 LED	2.5 m	1.7 mm

have achieved millimeter-level positioning accuracy with two cameras. In the above works, a commercial smartphone camera or industrial camera is adopted as the VLP receiver, which can be simplified to a pinhole camera. Compared with pinhole camera, fish-eye camera can achieve a wider field of view (FOV). For this reason, fish-eye camera is adopted in [27] and [28] to increase the number of detectable LEDs and improve the positioning accuracy. Moreover, the comparison of the proposed scheme with other camera-based VLP approaches is shown in Table I, where the proposed positioning approach shows higher positioning accuracy.

Recently, some works have theoretically analyzed the positioning error of VLP algorithm. In [29], the Cramer-Rao lower bound (CRLB) is presented as the theoretical accuracy limitation of RSS algorithm. Work [30] derived the closed-form expression of the CRLB for RSS algorithm under non-line-of-sight (NLOS) propagation. Work [31] presented the hybrid CRLB of the RSS-based hybrid position and orientation estimation algorithm. Moreover, a closed-form error expression for the AOA algorithm is derived in [32]. The above works are all related to the PD-based VLP, while the positioning error analysis is rarely investigated for the camera-based VLP systems.

In this work, we propose an indoor 3D visible light positioning scheme based on multiple cameras. Specifically, the pinhole camera is adopted as the VLP receiver to improve the positioning accuracy, whereas the fish-eye camera is adopted as the receiver to increase the coverage area. The main contributions of this paper are summarized as follows:

- We propose a multiple-camera-based visible light positioning (MC-VLP) algorithm, which adopts multiple cameras to simultaneously capture the beacon signal transmitted by the LED mounted on the mobile terminal. After establishing linear equations of incident rays, the 3D coordinate of the LED can be obtained by least squares (LS) estimation.
- We conduct theoretical analysis on the positioning error of MC-VLP algorithm, which reveals the impact of system parameters (i.e., camera height, tilt angle, layout radius, and the number of cameras) on the positioning error. The tilt angle and layout radius are optimized to reduce the positioning error in the coverage area.

The remainder of the paper is organized as follows. Section II describes the system model. Section III introduces the details of MC-VLP algorithm. Section IV presents theoretical analysis on the positioning error. Simulation results are presented in Section V. Experimental results are given in Section VI. Finally, conclusion is drawn in Section VII. In addition, some key variables and their meanings are summarized in Table II.

II. SYSTEM MODEL

As shown in Fig. 1(a), the considered MC-VLP system can be classified as sever-based VLP. Assume that K cameras are installed on the ceiling, evenly distributed on a circle with center Q (In this paper, a circle with center Q is termed circle Q for simplicity). Moreover, the K cameras are tilted towards center Q to increase the area of overlapping region. Once the LED to be located is captured by all the K cameras simultaneously, the positioning can be processed on the server to estimate the 3D world coordinate of the LED. The coordinate of LED can be fed back to the client terminal.

We establish two types of coordinate systems for the camera-based VLP, namely the 3D world coordinate system (WCS) and 3D camera coordinate system (CCS). The 3D WCS is $O-XYZ$, where XOY plane is the horizontal plane, and Z axis direction is vertically downward. The 3D CCS of camera k is CCS$_k$: $O_k - X_kY_kZ_k$, $k \in \mathcal{K} \triangleq \{1, 2, \cdots, K\}$, where origin O_k is the optical center of camera k, Y_k axis is tangent to circle Q, and Z_k axis is the optical axis of camera k. Let $\mathbf{b}_k^c = [x_k^c, y_k^c, z_k^c]^{\mathrm{T}}$ and $\mathbf{b}^w = [x^w, y^w, z^w]^{\mathrm{T}}$ denote the coordinates of the same point in the CCS$_k$ and WCS, respectively. The transformation from CCS$_k$ to WCS can be expressed as

$$\mathbf{b}^w = \mathbf{R}_k\mathbf{b}_k^c + \mathbf{t}_k,\ k \in \mathcal{K}, \tag{1}$$

where $\mathbf{t}_k$ and $\mathbf{R}_k$ are the 3×1 translation vector and 3×3 rotation matrix form CCS$_k$ to WCS, respectively. Denote the coordinate of point O_k in the WCS as $(x_{o_k}^w, y_{o_k}^w, z_{o_k}^w)$, $k \in \mathcal{K}$. Noting that point O_k is the origin of CCS$_k$, the translation vector $\mathbf{t}_k$ from CCS$_k$ to WCS is given by

$$\mathbf{t}_k = \left[x_{o_k}^w, y_{o_k}^w, z_{o_k}^w\right]^{\mathrm{T}},\ k \in \mathcal{K}. \tag{2}$$

Then, we consider the 3D rotation from CCS$_k$ to WCS. Let α_k, β_k, and γ_k denote the rotation angles around X_k, Y_k, and Z_k axes from CCS$_k$ to WCS in the counter-clockwise direction when looking downwards the origin O_k, respectively. The rotation matrices around X_k, Y_k, and Z_k axes can be expressed as

$$\mathbf{L}_x(\alpha_k) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha_k & \sin\alpha_k \\ 0 & -\sin\alpha_k & \cos\alpha_k \end{bmatrix}, \tag{3}$$

$$\mathbf{L}_y(\beta_k) = \begin{bmatrix} \cos\beta_k & 0 & -\sin\beta_k \\ 0 & 1 & 0 \\ \sin\beta_k & 0 & \cos\beta_k \end{bmatrix}, \tag{4}$$

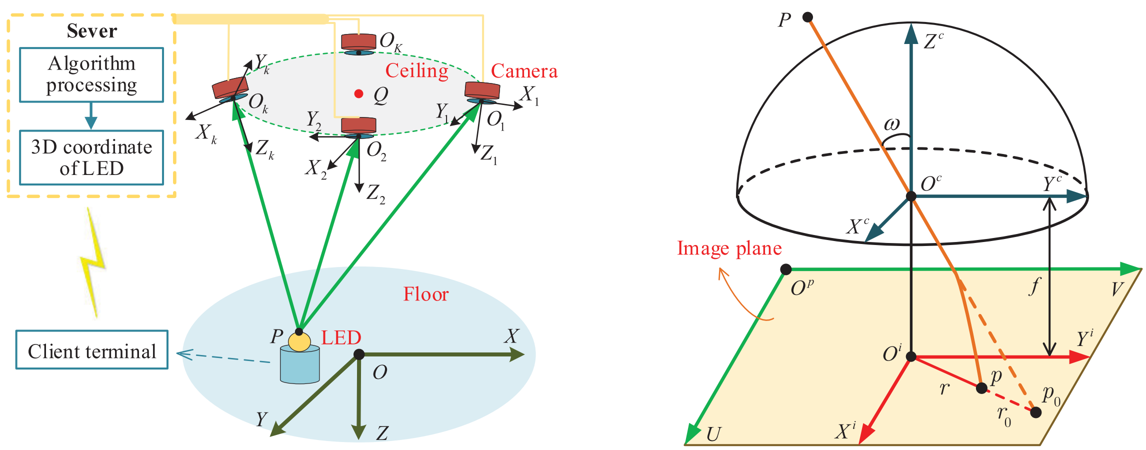
$$\mathbf{L}_z(\gamma_k) = \begin{bmatrix} \cos\gamma_k & \sin\gamma_k & 0 \\ -\sin\gamma_k & \cos\gamma_k & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{5}$$

Multiplying the above three matrices in the rotation order, rotation matrix $\mathbf{R}_k$ from CCS$_k$ to WCS can be expressed as

$$\mathbf{R}_k = \mathbf{L}_x(\alpha_k)\mathbf{L}_y(\beta_k)\mathbf{L}_z(\gamma_k),\ k \in \mathcal{K}. \tag{6}$$

TABLE II
SUMMARY OF KEY VARIABLES

Variable	Meaning	Variable	Meaning
f	Focal length of camera lens	K	The number of cameras
$\mathbf{R}$	Rotation matrix	R	Layout radius of cameras
$\mathbf{t}$	Translation vector	H	Camera height
(x^w, y^w, z^w)	Coordinate of point in WCS	θ	Tilt angle of camera
(x^c, y^c, z^c)	Coordinate of point in CCS	ω	Angle of incident ray
(x^i, y^i)	Coordinate of point in ICS	$\mathbf{W}$	Direction vector of incident ray
(u, v)	Coordinate of point in PCS	$(s, t, 0)$	Coordinate of LED in WCS



(a) (b)

Fig. 1. (a) Illustration of the considered MC-VLP system. (b) Illustration of camera model (The projection of point P on the image plane is point p_0 for a pinhole camera and point p for a fish-eye camera.)

Commonly used cameras include pinhole camera and fish-eye camera, whose projection models are shown in Fig. 1(b). Three coordinate systems are established for the cameras, namely 3D CCS $O^c - X^cY^cZ^c$, 2D image coordinate system (ICS) $O^i - X^iY^i$, and 2D pixel coordinate system (PCS) $O^p - UV$. Origin O^c of CCS is the optical center of camera, origin O^i of ICS is the principal point, and origin O^p of PCS is in the upper left corner of the image plane. Note that O^c and O^i are both on the optical axis; the distance between O^c and O^i is the focal length f; and the coordinate of O^i in the PCS is denoted as (u_0, v_0). In addition, X^c, X^i, and U axes are parallel to each other; Y^c, Y^i, and V axes are also parallel to each other.

Let (u, v) and (x^i, y^i) denote the coordinates of the same point in the PCS and ICS, respectively. The transformation from PCS to ICS can be expressed as

$$\begin{cases} x^i &= (u - u_0)d_x, \\ y^i &= (v - v_0)d_y, \end{cases} \tag{7}$$

where d_x and d_y are the physical sizes of each pixel in X^i axis and Y^i axis, respectively.

A. Pinhole Camera Model

As shown in Fig. 1(b), the projection of point P on the image plane is point p_0 for a pinhole camera, and the distance between image point p_0 and principal point O^i is r_0. Denote the coordinate of point P in the CCS by (x^c, y^c, z^c), and the coordinate of point p_0 in the ICS by $(x^i_{p_0}, y^i_{p_0})$. The transformation from CCS to ICS can be expressed as

$$\begin{cases} x^i_{p_0} &= f\dfrac{x^c}{z^c}, \\ y^i_{p_0} &= f\dfrac{y^c}{z^c}. \end{cases} \tag{8}$$

Furthermore, the perspective projection of a pinhole camera can be described by $r_0 = f\tan\omega$, where ω is the angle between optical axis Z^c and the incident ray.

B. Fish-Eye Camera Model

For a fish-eye camera, the light ray bends its direction after penetrating the lens. The projection of point P on the image plane is point p, the distance between image point p and principal point O^i is r. Fish-eye lenses are usually designed to obey one of the following projection models [33]:

$$\begin{cases} r = 2f\tan(\omega/2) & \text{(stereographic projection)}, \\ r = f\omega & \text{(equidistance projection)}, \\ r = 2f\sin(\omega/2) & \text{(equisolid angle projection)}, \\ r = f\sin\omega & \text{(orthogonal projection)}. \end{cases} \tag{9}$$

Letting $\mathcal{F}$ be the function of incident angle ω, the fish-eye camera model can be expressed as

$$r = \mathcal{F}(\omega). \tag{10}$$

III. MC-VLP ALGORITHM

In this section, we proposed a novel MC-VLP algorithm. The proposed MC-VLP algorithm contains three steps: 1) The direction vectors of incident rays are calculated according to the images captured by the cameras; 2) The linear equations of incident rays are established using the direction vectors of incident rays; 3) The 3D coordinate of the LED is obtained from the established equations using LS estimation.

A. Direction Vectors of Incident Rays

In Fig. 1(a), point P is the position of the LED to be located. After the LED is captured by all the K cameras, the projection of the LED on the image plane of camera k is denoted as point p_k, $k \in \mathcal{K}$. The coordinate of point p_k in the PCS is denoted as (u_{p_k}, v_{p_k}), which can be obtained by image recognition. Then, the coordinate of point p_k in the ICS, denoted by $(x^i_{p_k}, y^i_{p_k})$, can be obtained according to Eq. (7).

For the pinhole camera, the direction of the incident ray from point P to point O_k in the CCS$_k$ can be expressed as

$$\mathbf{W}^c_k = \left[x^i_{p_k}, y^i_{p_k}, -f_k\right]^{\mathrm{T}}, \ k \in \mathcal{K}, \tag{11}$$

where f_k is the focal length of camera k.

For the fish-eye camera, the distance between image point p_k and the principal point can be calculated as

$$r_k = \sqrt{\left(x^i_{p_k}\right)^2 + \left(y^i_{p_k}\right)^2}, \ k \in \mathcal{K}. \tag{12}$$

Letting $\mathcal{F}^{-1}$ be the inverse function of $\mathcal{F}$, the incident angle ω_k can be expressed as

$$\omega_k = \mathcal{F}^{-1}(r_k), \ k \in \mathcal{K}. \tag{13}$$

In the CCS$_k$, the direction of the incident ray from point P to point O_k can be expressed as

$$\mathbf{W}^c_k = \left[\frac{x^i_{p_k} f_k \tan\omega_k}{r_k}, \frac{y^i_{p_k} f_k \tan\omega_k}{r_k}, -f_k\right]^{\mathrm{T}}, \ k \in \mathcal{K}. \tag{14}$$

To sum up, the direction vectors of incident rays for pinhole camera and fish-eye camera are given in Eq. (11) and Eq. (14), respectively. Furthermore, based on rotation matrix $\mathbf{R}_k$, vector $\mathbf{W}^c_k$ in CCS$_k$ can be transformed into the following vector in the WCS:

$$\mathbf{W}^w_k = \left[x^w_{p_k},\, y^w_{p_k},\, z^w_{p_k}\right]^{\mathrm{T}} = \mathbf{R}_k \mathbf{W}^c_k, \ k \in \mathcal{K}. \tag{15}$$

To facilitate subsequent calculations, $\mathbf{W}^w_k$ can be normalized as

$$\bar{\mathbf{W}}_k = \left[\bar{x}^w_{p_k},\, \bar{y}^w_{p_k},\, 1\right]^{\mathrm{T}} = \left[x^w_{p_k}/z^w_{p_k},\, y^w_{p_k}/z^w_{p_k},\, 1\right]^{\mathrm{T}}, \ k \in \mathcal{K}. \tag{16}$$

B. Equation of Incident Ray

Since the K cameras are evenly distributed on circle Q, point Q is the geometric center of the K cameras as shown in Fig. 1(a). Using the coordinates of points $O_k, k \in \mathcal{K}$, the coordinate of point Q in the WCS can be expressed as

$$(x^w_q, y^w_q, z^w_q) = \left(\frac{1}{K}\sum_{k=1}^{K} x^w_{o_k}, \frac{1}{K}\sum_{k=1}^{K} y^w_{o_k}, \frac{1}{K}\sum_{k=1}^{K} z^w_{o_k}\right). \tag{17}$$

Theorem 1: In the WCS, given point $O_k = (x^w_{o_k}, y^w_{o_k}, z^w_{o_k})$, point $Q = (x^w_q, y^w_q, z^w_q)$, and the direction vector of incident ray $\bar{\mathbf{W}}_k = \left[\bar{x}^w_{p_k}, \bar{y}^w_{p_k}, 1\right]^{\mathrm{T}}$, the equation of the incident ray is given by

$$\begin{cases} A^r_k x + B^r_k y + C^r_k z = E^r_k, \ k \in \mathcal{K}, \\ A^t_k x + B^t_k y + C^t_k z = E^t_k, \ k \in \mathcal{K}, \end{cases} \tag{18}$$

where

$$\begin{cases} A^r_k = y^w_{o_k} - y^w_q, \\ B^r_k = x^w_q - x^w_{o_k}, \\ C^r_k = \bar{y}^w_{p_k}(x^w_{o_k} - x^w_q) - \bar{x}^w_{p_k}(y^w_{o_k} - y^w_q), \\ E^r_k = x^w_{o_k} A^r_k + y^w_{o_k} B^r_k + z^w_{o_k} C^r_k, \end{cases} \tag{19}$$

$$\begin{cases} A^t_k = x^w_{o_k} - x^w_q, \\ B^t_k = y^w_{o_k} - y^w_q, \\ C^t_k = \bar{y}^w_{p_k}(y^w_q - y^w_{o_k}) - \bar{x}^w_{p_k}(x^w_{o_k} - x^w_q), \\ E^t_k = x^w_{o_k} A^t_k + y^w_{o_k} B^t_k + z^w_{o_k} C^t_k. \end{cases} \tag{20}$$

Proof: Please refer to Appendix A. □

C. Position Estimation by LS Method

With K cameras in the MC-VLP system, there are K direction vectors of the incident rays whose intersection is point P. Based on the equation of incident ray in Eq. (18), we obtain $2K$ linear equations, which can be expressed in a matrix form as

$$\begin{bmatrix} A^r_1 & B^r_1 & C^r_1 \\ \vdots & \vdots & \vdots \\ A^r_K & B^r_K & C^r_K \\ A^t_1 & B^t_1 & C^t_1 \\ \vdots & \vdots & \vdots \\ A^t_K & B^t_K & C^t_K \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} E^r_1 \\ \vdots \\ E^r_K \\ E^t_1 \\ \vdots \\ E^t_K \end{bmatrix}. \tag{21}$$

Equation (21) can be denoted as $\mathbf{FX} = \mathbf{E}$, where $\mathbf{F}$ is a $2K \times 3$ matrix, $\mathbf{E}$ is a $2K \times 1$ vector, and $\mathbf{X}$ is a 3×1 unknown vector. The LS solution is given by

$$\hat{\mathbf{X}} = \left(\mathbf{F}^{\mathrm{T}}\mathbf{F}\right)^{-1}\mathbf{F}^{\mathrm{T}}\mathbf{E}, \tag{22}$$

where $\hat{\mathbf{X}}$ is the estimate of $\mathbf{X}$. Consequently, we obtain the 3D coordinate of LED.

Remark: The MC-VLP algorithm can be generalized to the case where the K cameras are not evenly distributed on a circle. More specifically, we require the coordinates $(O_k, k \in \mathcal{K})$ and postures $(\mathbf{R}_k, k \in \mathcal{K})$ of the K cameras as priori information. When the K cameras follow a non-circular layout, the coordinate of point Q in *Theorem 1* can still be calculated.

IV. A Theoretical Analysis on Positioning Error

In order to evaluate the performance of MC-VLP algorithm, we carry out a theoretical analysis on the positioning error considering the image noise on image sensor, including the positioning error along the X-axis, Y-axis, and Z-axis. To reduce the positioning error in the coverage area, we formulate an optimization problem with respect to the tilt angle and layout radius.

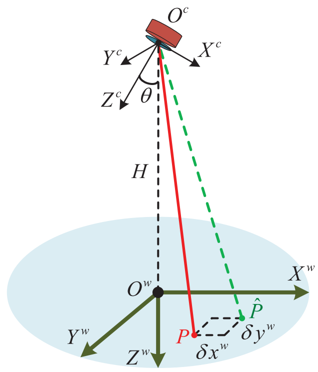


Fig. 2. Illustration of the observation error of camera.

A. Observation Error of Camera

Figure 2 illustrates the observation error of a camera, where $O^w - X^wY^wZ^w$ is the world coordinate system (WCS), and $O^c - X^cY^cZ^c$ is the camera coordinate system (CCS). In particular, the origin O^c is on the Z^w axis; the distance between O^c and O^w is height H; and the angle between Z^c axis and Z^w axis is θ. In addition, Y^c axis and Y^w axis are parallel to each other, such that $X^wO^wZ^w$ plane and $X^cO^cZ^c$ plane are the same and vertical to the horizontal plane.

The LED to be located is on the horizontal plane, and its real coordinate in the WCS is denoted as $P = (x^w, y^w, 0)$. The incident ray from the LED to the camera is represented by the red solid line. Due to the image noise on image sensor, the direction of the incident ray actually observed by the camera is represented by the green dotted line, and its intersection with the horizontal plane is denoted as $\hat{P} = (\hat{x}^w, \hat{y}^w, 0)$ in the WCS. The difference between points P and $\hat{P}$ is termed the observation error of camera, which can be expressed as

$$\begin{cases} \delta x^w = \hat{x}^w - x^w, \\ \delta y^w = \hat{y}^w - y^w, \end{cases} \tag{23}$$

where δx^w and δy^w are the observation errors along X^w axis and Y^w axis, respectively.

Let $p = (x^i, y^i)$ denote the theoretical projection of point P in the ICS $O^i - X^iY^i$, and $\hat{p} = (\hat{x}^i, \hat{y}^i)$ denote the estimated position of point p. The image noises along X^i axis and Y^i axis can be expressed as

$$\begin{cases} \delta x^i = \hat{x}^i - x^i, & \delta x^i \sim N(0, \sigma_x^2), \\ \delta y^i = \hat{y}^i - y^i, & \delta y^i \sim N(0, \sigma_y^2), \end{cases} \tag{24}$$

where δx^i and δy^i obey Gaussian distribution.

As shown in Fig. 2, the translation vector $\mathbf{t}$ from CCS to WCS is $\mathbf{t} = [0, 0, -H]^{\mathrm{T}}$. According to Eq. (6), rotation matrix $\mathbf{R}$ from CCS to WCS is given by $\mathbf{R} = \mathbf{L}_x(0)\mathbf{L}_y(\theta)\mathbf{L}_z(0) = \mathbf{L}_y(\theta)$. Thus, the real coordinate of point P can be transformed from WCS into CCS and expressed as

$$\begin{bmatrix} x^c \\ y^c \\ z^c \end{bmatrix} = \mathbf{R}^{\mathrm{T}} \left\{ \begin{bmatrix} x^w \\ y^w \\ 0 \end{bmatrix} - \mathbf{t} \right\} = \begin{bmatrix} H\sin\theta + x^w\cos\theta \\ y^w \\ H\cos\theta - x^w\sin\theta \end{bmatrix}. \tag{25}$$

Theorem 2: Consider a pinhole camera with focal length f and tilt angle θ. Denote the real coordinate of the observation point in the CCS by (x^c, y^c, z^c), and the vertical distance between the observation point and pinhole camera by H. The relationship between the image noise vector and the observation error vector is given by

$$\begin{bmatrix} \delta x^w \\ \delta y^w \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{(z^c)^2}{fH} & 0 \\ -\frac{y^c z^c \sin\theta}{fH} & \frac{z^c}{f} \end{bmatrix}}_{\mathbf{Q}_1(x^c, y^c, z^c, H, \theta, f)} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix}. \tag{26}$$

Proof: Please refer to Appendix B. □

Specially, when the optical axis of pinhole camera is vertically downward, we have $\theta = 0$, $z^c = H$, such that $\mathbf{Q}_1(\cdot) = (H/f)\,\mathbf{I}_{2\times 2}$.

Theorem 3: Consider a fish-eye camera with model function $\mathcal{F}$ and tilt angle θ. Denote the real coordinate of the observation point in the CCS by (x^c, y^c, z^c), and the vertical distance between the observation point and fish-eye camera by H. The relationship between the image noise vector and the observation error vector is given by Eq. (27), shown at the bottom of the next page, where $\omega = \arctan\frac{\sqrt{(x^c)^2+(y^c)^2}}{z^c}$, $\xi = (x^c)^2 + (y^c)^2$, and $\eta = x^c z^c \xi^{-1}\cos\theta + \sin\theta$.

Proof: Please refer to Appendix C. □

B. Expression of Positioning Error

Figure 3(a) illustrates the observation error of camera k in the MC-VLP system, where circle Q is parallel to the XOY plane, the radius of circle Q is R, and the coordinate of point Q is $(0, 0, -H)$ in the WCS. Assume that the K cameras are evenly distributed on circle Q, and the cameras are located in a clockwise manner starting from $(R, 0, -H)$. Then, the angle between X axis and QO_k can be expressed as

$$\psi_k = \frac{2\pi}{K}(k-1),\ k \in \mathcal{K}. \tag{28}$$

The real coordinate of the LED is denoted as $P = (s, t, 0)$ in the WCS $O - XYZ$. After point P is captured by camera k, the intersection of observed incident ray and horizontal plane is $\hat{P}_k = (\hat{s}_k, \hat{t}_k, 0)$ in the presence of image noise on image sensor. The difference between points P and $\hat{P}_k$ is the observation error of camera k.

Figure 3(b) shows the observation error in two coordinate systems, where coordinate system $O - X_k'Y_k'Z$ is obtained by rotating coordinate system $O - XYZ$ with an angle ψ_k about the Z axis in the clockwise direction. Coordinate system $O - X_k'Y_k'Z$ is related to the camera k, X_k' axis is parallel to QO_k, and Y_k' axis is parallel to Y_k axis. In coordinate system $O - XYZ$, the observation error along X axis and Y axis are denoted as δs_k and δt_k, respectively. We have

$$\begin{cases} \delta s_k = \hat{s}_k - s, & k \in \mathcal{K}, \\ \delta t_k = \hat{t}_k - t, & k \in \mathcal{K}. \end{cases} \tag{29}$$

In coordinate system $O - X_k'Y_k'Z$, the observation error along X_k' axis and Y_k' axis are denoted as δg_k and δq_k, respectively. In addition, δg_k and δq_k are also called the radial component and tangent component of the observation error, respectively.

In Eq. (26) and Eq. (27), (x^c, y^c, z^c) is the real coordinate of the observation point P in the CCS. Note that the relative

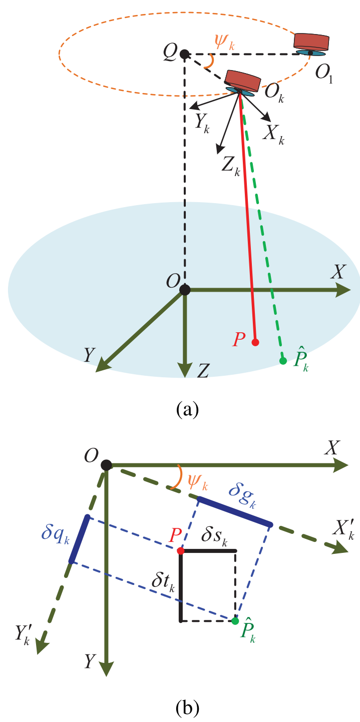


Fig. 3. (a) Illustration of the observation error of camera k in the MC-VLP system. (b) Illustration of the observation error in two coordinate systems (the Z axis is perpendicular to the plane of the paper and pointing inward).

position between the WCS and CCS in Fig. 2 and Fig. 3(a) are different, and we need to derive the real coordinate of the observation point P in the CCS $O_k - X_kY_kZ_k$. As shown in Fig. 3(a), the translation vector $\mathbf{t}_k$ from coordinate system $O_k - X_kY_kZ_k$ to coordinate system $O - XYZ$ can be given by $\mathbf{t}_k = \left[R\cos\psi_k, R\sin\psi_k, -H\right]^{\mathrm{T}}$. According to Eq. (6), the rotation matrix $\mathbf{R}_k$ from coordinate system $O - XYZ$ to coordinate system $O_k - X_kY_kZ_k$ can be given by $\mathbf{R}_k = \mathbf{L}_x(0)\mathbf{L}_y(-\theta)\mathbf{L}_z(\psi_k)$. Thus, the real coordinate of point P can be transformed from coordinate system $O - XYZ$ into coordinate system $O_k - X_kY_kZ_k$ and expressed as

$$\begin{bmatrix} x_k^c \\ y_k^c \\ z_k^c \end{bmatrix} = \mathbf{R}_k \left\{ \begin{bmatrix} s \\ t \\ 0 \end{bmatrix} - \mathbf{t}_k \right\} = \begin{bmatrix} \cos\theta(s\cos\psi_k + t\sin\psi_k - R) + H\sin\theta \\ -s\sin\psi_k + t\cos\psi_k \\ -\sin\theta(s\cos\psi_k + t\sin\psi_k - R) + H\cos\theta \end{bmatrix}. \tag{30}$$

According to Theorem 2 and Theorem 3, the radial component δg_k and tangent component δq_k of observation error is given by

$$\begin{bmatrix} \delta g_k \\ \delta q_k \end{bmatrix} = \begin{cases} \mathbf{Q}_1(x_k^c, y_k^c, z_k^c, H, \theta, f)\left[\delta x^i,\, \delta y^i\right]^{\mathrm{T}}, \\ \qquad \text{for pinhole camera}, \\ \mathbf{Q}_2(x_k^c, y_k^c, z_k^c, H, \theta, \mathcal{F})\left[\delta x^i,\, \delta y^i\right]^{\mathrm{T}}, \\ \qquad \text{for fish-eye camera}. \end{cases} \tag{31}$$

To facilitate subsequent derivations, Eq. (31) is denoted as

$$\begin{bmatrix} \delta g_k \\ \delta q_k \end{bmatrix} = \begin{bmatrix} \bar{A}_k & \bar{B}_k \\ \bar{C}_k & \bar{D}_k \end{bmatrix} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix}. \tag{32}$$

According to Fig. 3(b), we can obtain the transformation relationship among these components of observation error. The X-axis component δs_k and Y-axis component δt_k of the observation error is given by

$$\begin{bmatrix} \delta s_k \\ \delta t_k \end{bmatrix} = \begin{bmatrix} \cos\psi_k & -\sin\psi_k \\ \sin\psi_k & \cos\psi_k \end{bmatrix} \begin{bmatrix} \delta g_k \\ \delta q_k \end{bmatrix}. \tag{33}$$

Substituting Eq. (32) into Eq. (33), we have

$$\begin{aligned} \begin{bmatrix} \delta s_k \\ \delta t_k \end{bmatrix} &= \begin{bmatrix} \bar{M}_k & \bar{N}_k \\ \bar{U}_k & \bar{V}_k \end{bmatrix} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix} \\ &= \begin{bmatrix} \bar{A}_k\cos\psi_k - \bar{C}_k\sin\psi_k & \bar{B}_k\cos\psi_k - \bar{D}_k\sin\psi_k \\ \bar{A}_k\sin\psi_k + \bar{C}_k\cos\psi_k & \bar{B}_k\sin\psi_k + \bar{D}_k\cos\psi_k \end{bmatrix} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix}. \end{aligned} \tag{34}$$

Theorem 4: Denote the number of cameras by K $(K \geq 2)$, the tilt angle of cameras by θ, the layout radius of cameras by R, the real coordinate of the positioning point in the WCS by $(s, t, 0)$, and the vertical distance between the positioning point and camera by H. The positioning error of the MC-VLP algorithm along the X-axis, Y-axis, and Z-axis is given by

$$\begin{cases} \Delta X = \dfrac{R^2\tau_s - R\tau_s\tau_r/K}{KR^2 - 2R\tau_r + \mathcal{M}} - s, \\ \Delta Y = \dfrac{R^2\tau_t - R\tau_t\tau_r/K}{KR^2 - 2R\tau_r + \mathcal{M}} - t, \\ \Delta Z = \dfrac{H(R\tau_r - \mathcal{M})}{KR^2 - 2R\tau_r + \mathcal{M}}, \end{cases} \tag{35}$$

where $\tau_s = Ks + \sum\limits_{k=1}^{K}\delta s_k$, $\tau_t = Kt + \sum\limits_{k=1}^{K}\delta t_k$, $\tau_r = \sum\limits_{k=1}^{K}\delta g_k$, and $\mathcal{M}$ is given by

$$\mathcal{M} = \sum_{k=1}^{K}\left[(\delta s_k)^2 + (\delta t_k)^2\right] - \frac{1}{K}\left(\sum_{k=1}^{K}\delta s_k\right)^2$$

$$\begin{bmatrix} \delta x^w \\ \delta y^w \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\left[\xi+(z^c)^2\right](x^c)^2}{H\xi\mathcal{F}'(\omega)} + \frac{(y^c)^2 z^c \xi^{-\frac{1}{2}}}{H\mathcal{F}(\omega)} & \frac{\left[\xi+(z^c)^2\right]x^c y^c}{H\xi\mathcal{F}'(\omega)} - \frac{x^c y^c z^c \xi^{-\frac{1}{2}}}{H\mathcal{F}(\omega)} \\ \frac{\left[\xi+(z^c)^2\right]x^c y^c \cos\theta}{H\xi\mathcal{F}'(\omega)} - \frac{y^c \eta \xi^{\frac{1}{2}}}{H\mathcal{F}(\omega)} & \frac{\left[\xi+(z^c)^2\right](y^c)^2 \cos\theta}{H\xi\mathcal{F}'(\omega)} + \frac{x^c \eta \xi^{\frac{1}{2}}}{H\mathcal{F}(\omega)} \end{bmatrix}}_{\mathbf{Q}_2(x^c, y^c, z^c, H, \theta, \mathcal{F})} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix}. \tag{27}$$

$$-\frac{1}{K}\left(\sum_{k=1}^{K}\delta t_k\right)^2. \tag{36}$$

With $KR \gg 2\tau_r$ and high-order small quantities (e.g., $\mathcal{M}$ and $\tau_r \sum_{k=1}^{K} \delta s_k$) neglected, Eq. (35) can be approximated as

$$\begin{cases} \Delta X \approx \dfrac{s}{KR}\displaystyle\sum_{k=1}^{K}\delta g_k + \dfrac{1}{K}\sum_{k=1}^{K}\delta s_k, \\ \Delta Y \approx \dfrac{t}{KR}\displaystyle\sum_{k=1}^{K}\delta g_k + \dfrac{1}{K}\sum_{k=1}^{K}\delta t_k, \\ \Delta Z \approx \dfrac{H}{KR}\displaystyle\sum_{k=1}^{K}\delta g_k, \end{cases} \tag{37}$$

where δg_k, δs_k, and δt_k are given by Eq. (31) and Eq. (33).

Proof: Please refer to Appendix D. □

C. Optimization of Tilt Angle and Layout Radius

The accuracy of the MC-VLP algorithm is quantified by the positioning error. Specifically, the 3D positioning error is defined as

$$\Delta D = \sqrt{(\Delta X)^2 + (\Delta Y)^2 + (\Delta Z)^2}. \tag{38}$$

Theorem 5: Following the same parameters and approximation results as those in Theorem 4, the positioning error metric of the MC-VLP algorithm is defined as the expectation of $(\Delta D)^2$, given by

$$\begin{aligned} &\mathbb{E}[(\Delta D)^2] \\ &= E_p(s,t,H,\theta,R,K) \\ &= \frac{s^2+t^2+H^2}{K^2R^2}\left[\sigma_x^2\sum_{k=1}^{K}\left(\bar{A}_k\right)^2 + \sigma_y^2\sum_{k=1}^{K}\left(\bar{B}_k\right)^2\right] \\ &\quad + \frac{\sigma_x^2}{K^2}\sum_{k=1}^{K}\left[(\bar{A}_k)^2+(\bar{C}_k)^2\right] + \frac{\sigma_y^2}{K^2}\sum_{k=1}^{K}\left[(\bar{B}_k)^2+(\bar{D}_k)^2\right] \\ &\quad + \frac{2s}{K^2R}\left[\sigma_x^2\sum_{k=1}^{K}\left(\bar{A}_k\bar{M}_k\right) + \sigma_y^2\sum_{k=1}^{K}\left(\bar{B}_k\bar{N}_k\right)\right] \\ &\quad + \frac{2t}{K^2R}\left[\sigma_x^2\sum_{k=1}^{K}\left(\bar{A}_k\bar{U}_k\right) + \sigma_y^2\sum_{k=1}^{K}\left(\bar{B}_k\bar{V}_k\right)\right], \end{aligned} \tag{39}$$

where $\bar{A}_k, \bar{B}_k, \bar{C}_k$ and $\bar{D}_k$ are given by Eq. (32); $\bar{M}_k, \bar{N}_k, \bar{U}_k$ and $\bar{V}_k$ are given by Eq. (34); and σ_x^2 and σ_y^2 are the variances of image noises δx^i and δy^i, respectively.

Proof: Please refer to Appendix E. □

Assume that the positioning points are evenly distributed in the coverage area, whose real coordinates in the WCS are denoted by $(s_i, t_i, 0)$, $i \in \{1, 2, \cdots, I\}$. To improve the positioning accuracy in the coverage area, we minimize the mean E_p over all the I positioning points by optimizing the tilt angle θ and layout radius R, i.e.,

$$\min_{\theta,R}\frac{1}{I}\sum_{i=1}^{I}E_p(s_i,t_i,H,\theta,R,K), \tag{40}$$

where E_p is given by Theorem 5. In this work, we solve the optimization problem (40) by particle swarm optimization (PSO) algorithm to obtain the optimal tilt angle and layout radius.

V. Simulation Results and Discussions

Unless otherwise specified, the system parameters in the simulations are set as follows. The pinhole camera has focal length $f = 5$ mm, and pixel size $d_x = d_y = 5.5$ μm. The fish-eye camera obeys equidistance projection, with focal length $f = 1$ mm and pixel size $d_x = d_y = 2.2$ μm. In addition, all the cameras are mounted on the ceiling at height $H = 2$ m, and the LED to be located is distributed in horizontal plane. The image noises δx^i and δy^i are modeled as zero-mean Gaussian noises with standard deviations of 3 pixels, i.e., $\sigma_x = \sigma_y = 3d_x = 3d_y$.

A. Simulation and Theoretical Results of Positioning Error

In order to verify the accuracy of MC-VLP algorithm and the correctness of Theorem 4, we provide the simulation results of MC-VLP algorithm when considering the image noises, as well as the theoretical calculation results of Theorem 4. The detailed procedure of simulation is given as follows: 1) Set the parameters of positioning system and initialize the world coordinates of the points to be located. 2) Simulate the imaging process of cameras, and obtain the coordinates of the projected points on image plane. 3) Generate Gaussian random variables to simulate image noises. 4) Adopt MC-VLP algorithm to estimate the 3D coordinates of the target points.

Firstly, four pinhole cameras are adopted as the VLP receivers, with tilt angle $\theta = 7°$ and layout radius $R = 1$ m. As shown in Fig. 4 and Fig. 5, the simulation results are consistent with the theoretical results. Furthermore, the distributions of positioning errors along X-axis and Y-axis are symmetrical about axes $X = 0$ m and $Y = 0$ m, respectively. The smaller errors in the central area corresponds to the larger errors in the edge area. The distribution of positioning errors along Z-axis is uniform, and the distribution of 3D positioning errors is centrosymmetric.

Secondly, four fish-eye cameras are employed as the VLP receivers, with tilt angle $\theta = 7°$ and layout radius $R = 1$ m. As shown in Fig. 6 and Fig. 7, the simulation results are consistent with the theoretical results. Furthermore, the distributions of positioning errors along X-axis and Y-axis are symmetrical about axes $X = 0$ m and $Y = 0$ m, respectively. The smaller errors in the central area corresponds to the larger errors in the edge area. The distributions of positioning errors along Z-axis and 3D positioning errors are both centrosymmetric.

In the above simulation results, the number of cameras K, layout radius R and camera height H are determined. The distributions of positioning errors can be justified according to Eq. (37). The positioning error along X-axis is mainly determined by the X-axis coordinate, where larger X-axis coordinate leads to larger positioning error along X-axis. The positioning error along Y-axis follows a similar trend. However, the positioning error along Z-axis is mainly determined by the radial component of observation error, which does not

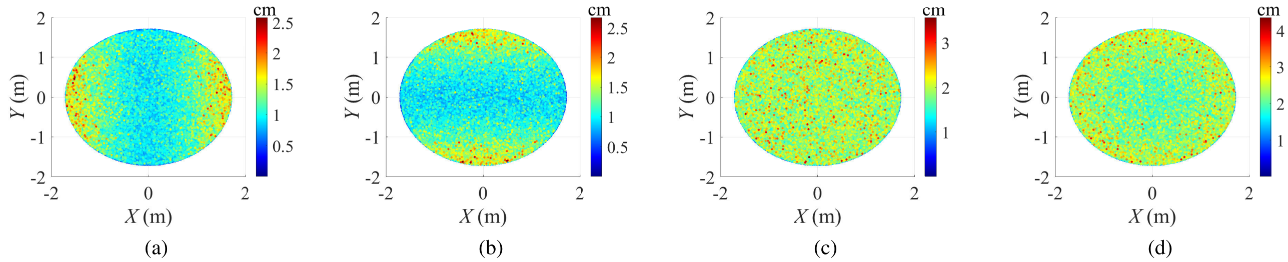


Fig. 4. The simulation results of positioning error with 4 pinhole cameras. (a) Positioning error along X-axis. (b) Positioning error along Y-axis. (c) Positioning error along Z-axis. (d) 3D positioning error.

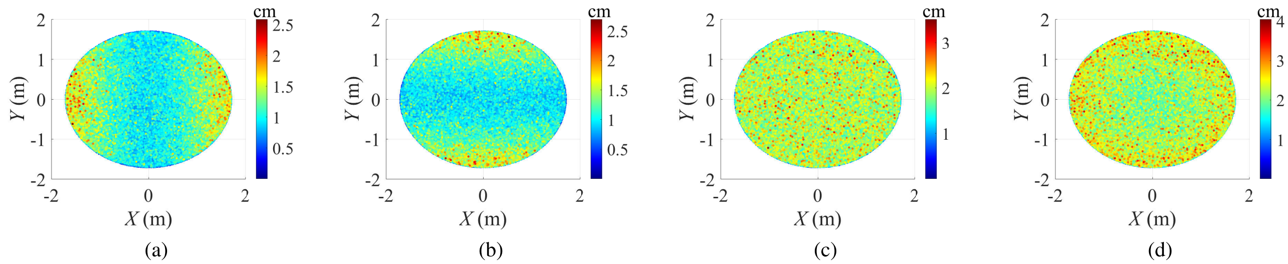


Fig. 5. The theoretical results of positioning error with 4 pinhole cameras. (a) Positioning error along X-axis. (b) Positioning error along Y-axis. (c) Positioning error along Z-axis. (d) 3D positioning error.

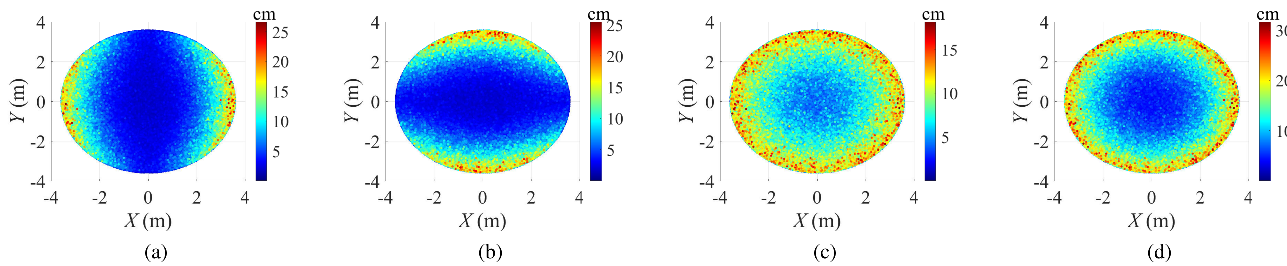


Fig. 6. The simulation results of positioning error with 4 fish-eye cameras. (a) Positioning error along X-axis. (b) Positioning error along Y-axis. (c) Positioning error along Z-axis. (d) 3D positioning error.

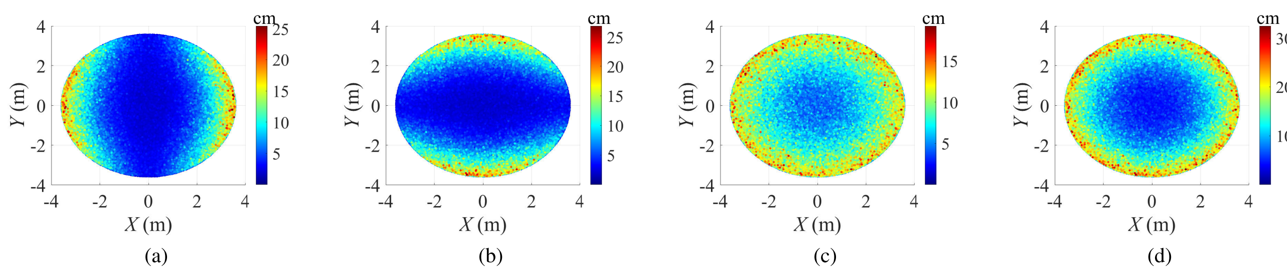


Fig. 7. The theoretical results of positioning error with 4 fish-eye cameras. (a) Positioning error along X-axis. (b) Positioning error along Y-axis. (c) Positioning error along Z-axis. (d) 3D positioning error.

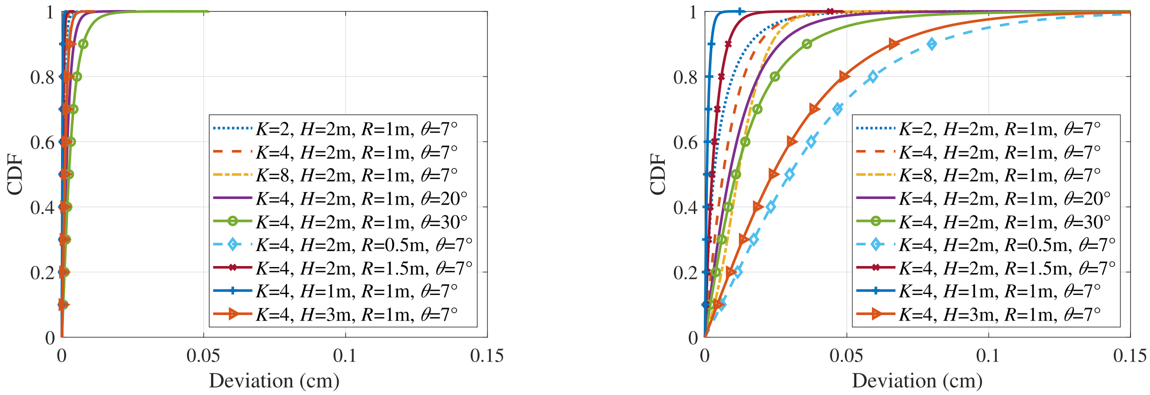
change significantly in the positioning area of the pinhole camera, but increases gradually from the central area to the edge area of the fish-eye camera. Moreover, compared with fish-eye camera receiver, pinhole camera receiver can reduce the positioning error with smaller coverage area.

To further explore the differences between simulation and theoretical results, we have calculated the deviations and drawn the CDF curves under the 9 considered system parameter settings. Moreover, we calculate the theoretical results by Eqs. (35) and (37) in Theorem 4, where Eq. (37) is an approximation of Eq. (35).

As shown in Fig. 8, when pinhole cameras are adopted as the VLP receivers, the deviations between simulation and theoretical results are very small. The deviations increase when adopting the approximate equation, but are still generally less than 0.15 cm under all the 9 system parameter settings. As shown in Fig. 9, when fish-eye cameras are adopted as the VLP receivers, the deviations are lower than 0.2 cm and 1 cm when using Eq. (35) and Eq. (37), respectively. Since the positioning error with fish-eye camera is significantly larger than that with pinhole camera, the larger deviations with fish-eye camera are acceptable.

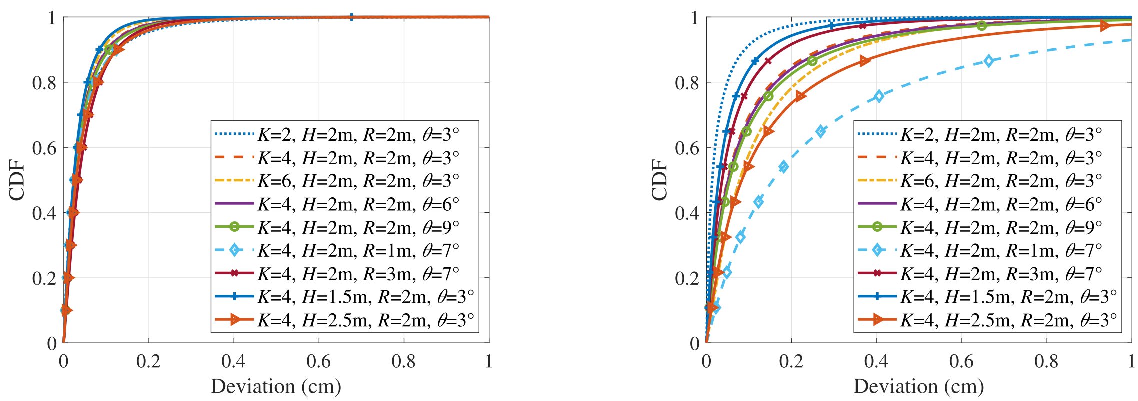
B. Effect of System Parameters on the Positioning Error

We explore the effect of system parameters on the positioning error, including the number of cameras K, tilt angle θ, layout radius R, and camera height H. In the simulation, we set the world coordinate of the positioning point as $(1, 1, 0)$. After the system parameters are determined, we can calculate the positioning errors ΔX, ΔY, and ΔZ by Theorem 4, and obtain the 3D positioning error ΔD.



(a) (b)

Fig. 8. Deviation between simulation and theoretical results with pinhole camera. (a) Theoretical results using Eq. (35). (b) Theoretical results using Eq. (37).



(a) (b)

Fig. 9. Deviation between simulation and theoretical results with fish-eye camera. (a) Theoretical results using Eq. (35). (b) Theoretical results using Eq. (37).

In particular, the 3D positioning error ΔD under the same point is calculated 10^6 times, and the 99.5% quantile of the 3D positioning errors is selected.

1) Effect of the Number of Cameras: The 3D positioning error versus the number of cameras under different projection models is shown in Fig. 10(a), with tilt angle $\theta = 7°$, layout radius $R = 0.5$ m, and camera height $H = 2$ m. It is shown that the 3D positioning error decreases with the number of cameras. In addition, the 3D positioning error of pinhole camera under perspective projection is significantly lower than those under other projections, and the positioning error of fish-eye camera increases with the lens distortion caused by the projection model.

2) Effect of the Tilt Angle: The 3D positioning error versus the tilt angle under different projection models is shown in Fig. 10(b), with the number of cameras $K = 4$, layout radius $R = 0.5$ m, and camera height $H = 2$ m. It is shown that the 3D positioning error of fish-eye camera under orthogonal projection and equisolid angle projection first decrease slightly and then increase gradually with the tilt angle, the 3D positioning error of fish-eye camera under equidistance projection remains almost constant, and the 3D positioning error of fish-eye camera under stereographic projection and pinhole camera under perspective projection first increase slightly and then decrease slightly. Specially, the 3D positioning error of fish-eye camera under orthogonal projection shows more drastic change with the tilt angle, since lens distortion caused by orthogonal projection is more significant than those caused by other projections.

3) Effect of the Layout Radius: The 3D positioning error versus the layout radius under different projection models is presented in Fig. 10(c), with the number of cameras $K = 4$, tilt angle $\theta = 15°$, and camera height $H = 2$ m. It is shown that the 3D positioning error decreases sharply as layout radius increases from 0.2 m to 1.1 m, and first decreases slightly and then increases slightly as layout radius increases from 1.1 m to 2.9 m. It implies that the layout radius should not be too small in order to guarantee a high positioning accuracy.

4) Effect of the Camera Height: The 3D positioning error versus the camera height under different projection models is shown in Fig. 10(d), with the number of cameras $K = 4$, tilt angle $\theta = 15°$, and layout radius $R = 1$ m. It is shown that the 3D positioning error increases gradually with the camera height. This is because large camera height leads to large observation error and further large 3D positioning error. In addition, the 3D positioning errors of fish-eye camera are close under four projection models.

Table III presents the monotonicity of 3D positioning error with respect to different parameters. In Table III, "↘" represents decreasing; "↗" represents increasing;

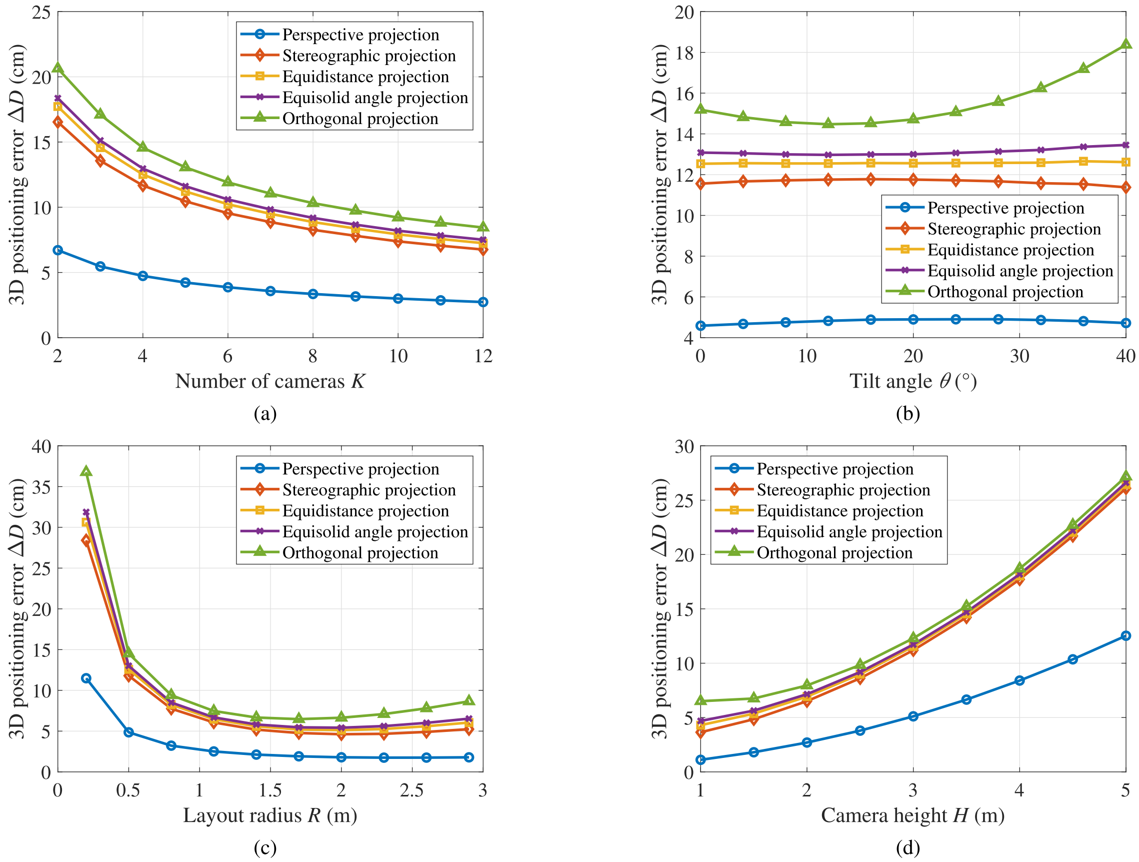


Fig. 10. (a) The 3D positioning error versus the number of cameras under different projection models. (b) The 3D positioning error versus the tilt angle under different projection models. (c) The 3D positioning error versus the layout radius under different projection models. (d) The 3D positioning error versus the camera height under different projection models.

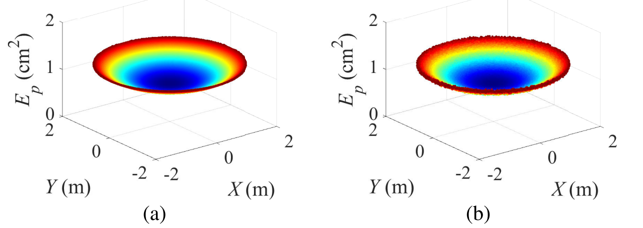


Fig. 11. (a) Theoretical results and (b) simulation results of E_p with 4 pinhole cameras.

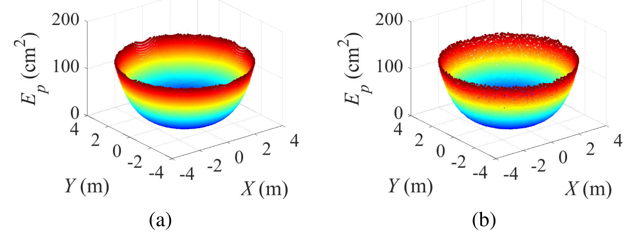


Fig. 12. (a) Theoretical results and (b) simulation results of E_p with 4 fish-eye cameras.

"↗ ↘" represents first increasing and then decreasing; "↘ ↗" represents first decreasing and then increasing.

C. Camera Layout and Tilt Angle Optimization

We verify the correctness of Theorem 5 by comparing the theoretical results and simulation results of E_p, with the number of cameras $K = 4$, the tilt angle $\theta = 7°$, layout radius $R = 1$ m, and camera height $H = 2$ m. As shown in Fig. 11(a) and Fig. 11(b), the theoretical results and simulation results fit well for pinhole camera, the distribution of E_p shows centrosymmetric, with lower value in the central area. As shown in Fig. 12(c) and Fig. 12(d), the theoretical results and simulation results also fit well for fish-eye camera, where larger E_p compared with pinhole camera implies larger positioning error.

Then, we investigate the improvement of positioning accuracy achieved by optimizing the tilt angle θ and layout radius R. We set the number of cameras $K = 4$, camera height $H = 2$ m, and the number of positioning points $I = 7668$. For the pinhole camera with perspective projection, we assume $\theta \in [0, 20]$, $R \in [0, 2]$ and that the positioning points are evenly distributed in a circular area with radius 2 m. For the fish-eye camera, we assume $\theta \in [0, 60]$, $R \in [0, 4]$ and that the positioning points are evenly distributed in a circular area with radius 4 m. In the PSO algorithm, we set the particle number to 20, the maximum iteration number to 20, the maximum limited velocity to 1, the inertia weight to 0.5, and the accelerate constants to $c_1 = c_2 = 1.5$.

Table IV presents the optimized tilt angle θ and layout radius R, as well as the mean 3D positioning error. In addition, the mean 3D positioning errors under four different sets of parameters are also given as the performance comparison benchmark. It is seen that the minimum mean 3D positioning error can be achieved by the optimal θ and R. Furthermore,

TABLE III
MONOTONICITY OF 3D POSITIONING ERROR WITH RESPECT TO DIFFERENT PARAMETERS

Projection model \ **Parameter**	**Number of cameras** K	**Tilt angle** θ (°)	**Layout radius** R (m)	**Camera height** H (m)
Perspective projection	↘	↗ ↘	↘ ↗	↗
Stereographic projection	↘	↗ ↘	↘ ↗	↗
Equidistance projection	↘	almost constant	↘ ↗	↗
Equisolid angle projection	↘	↘ ↗	↘ ↗	↗
Orthogonal projection	↘	↘ ↗	↘ ↗	↗

TABLE IV
OPTIMIZED PARAMETERS AND MEAN 3D POSITIONING ERROR

Projection model	**Optimized parameter**		**Mean 3D positioning error (cm)**				
	Tilt angle θ (°)	Layout radius R (m)	Optimal θ and R	$\theta = 0°$ $R = 1$m	$\theta = 0°$ $R = 2$m	$\theta = 20°$ $R = 1$m	$\theta = 20°$ $R = 2$m
Perspective projection	0	2	**0.56**	0.82	–	0.97	0.82
Stereographic projection	0	2.80	**2.42**	4.02	2.56	4.26	2.86
Equidistance projection	0	2.68	**3.16**	5.03	3.27	5.05	3.30
Equisolid angle projection	48.17	2.75	**3.36**	5.67	3.73	5.54	3.55
Orthogonal projection	49.88	3.18	**3.87**	9.49	6.90	8.74	4.70

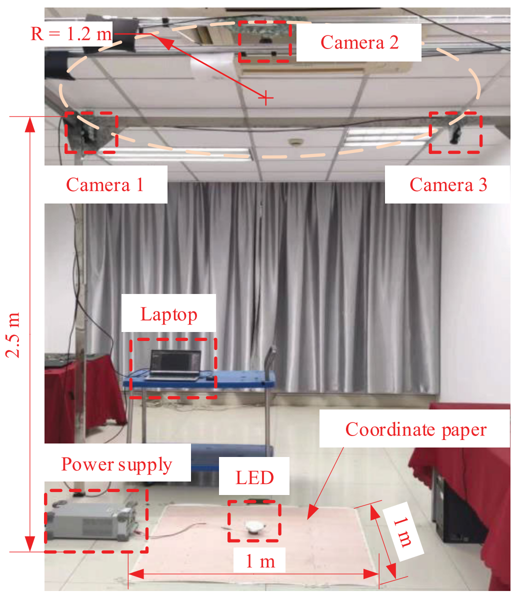


Fig. 13. Experimental platform.

for the pinhole camera of perspective projection, the mean 3D positioning error is in the millimeter level. For the fish-eye cameras of stereographic projection, equidistance projection, equisolid angle projection, and orthogonal projection, the mean 3D positioning errors are 2.42 cm, 3.16 cm, 3.36 cm, and 3.87 cm, respectively.

VI. EXPERIMENT AND RESULTS

A. Experimental Setup

Figure 13 shows the experimental platform, where three cameras are mounted on the ceiling at height 2.5 m, evenly distributed on a circle with a radius of 1.2 m, and tilted towards center at the same angle. The LED to be positioned is placed on the coordinate paper, and a laptop is employed to control the 3 cameras and process the MC-VLP algorithm. The experimental configurations are summarized in Table V.

Before the experiment, we conduct camera calibration using the method proposed in [34]. After capturing chessboard images with the 3 cameras simultaneously, we complete the camera calibration and obtain the world coordinates, rotation matrices, focal lengths, and lens distortion coefficients of the 3 cameras. The lens distortion coefficients are utilized to correct the images captured by the cameras, and the remaining parameters are adopted in the MC-VLP algorithm.

TABLE V
EXPERIMENTAL PARAMETERS

	Parameter	**Value**
Platform	Experimental area	1×1 m^2
	The number of cameras	3
	Layout radius of cameras	1.2 m
	Camera height	2.5 m
	Tilt angle of cameras	25 °
Camera	Model	MV-CE120-10UM
	Resolution	4000×3036
	Pixel size	1.85 µm
	Focal length	5 mm

In our experiment, a total of 36 test points with a spacing of 20 cm are selected in the experimental area of 1×1 m^2. Then, the LED is placed on each test point, and 3 cameras capture the image of LED. After calculating the 3D world coordinate of LED using the proposed MC-VLP algorithm, we quantify the positioning accuracy by the 3D positioning error.

B. Experimental Results

Figure 14(a) shows the distribution of 3D positioning errors, and Fig. 14(b) shows the corresponding cumulative distribution function (CDF). The 3D positioning errors range from 0.4 mm to 3.7 mm, with a mean value of 1.7 mm. The errors along X-axis range from 0 mm to 3.1 mm, with a mean value of 1.0 mm. The errors along Y-axis range from 0 mm to 2.8 mm, with a mean value of 1.2 mm. The errors along Z-axis range from 0 mm to 0.7 mm, with a mean value of 0.2 mm. Moreover, we also select two cameras (Camera 1 and Camera 2) to realize the positioning, and all the positioning results are summarized in Table VI.

It can be observed that the 3D positioning error with 3 cameras is lower than that with 2 cameras, which demonstrates our theoretical result that the 3D positioning error decreases with the number of cameras. Furthermore, the positioning errors along X-axis and Y-axis are very close for the VLP

TABLE VI
POSITIONING RESULTS

The number of cameras	2 cameras				3 cameras			
Positioning error	X-axis	Y-axis	Z-axis	3D	X-axis	Y-axis	Z-axis	3D
Max (mm)	3.3	3.6	5.9	6.8	3.1	2.8	0.7	**3.7**
Mean (mm)	1.0	1.3	1.8	2.6	1.0	1.2	0.2	**1.7**
90% Quantile (mm)	2.0	2.7	3.8	5.0	2.2	2.4	0.5	**3.0**

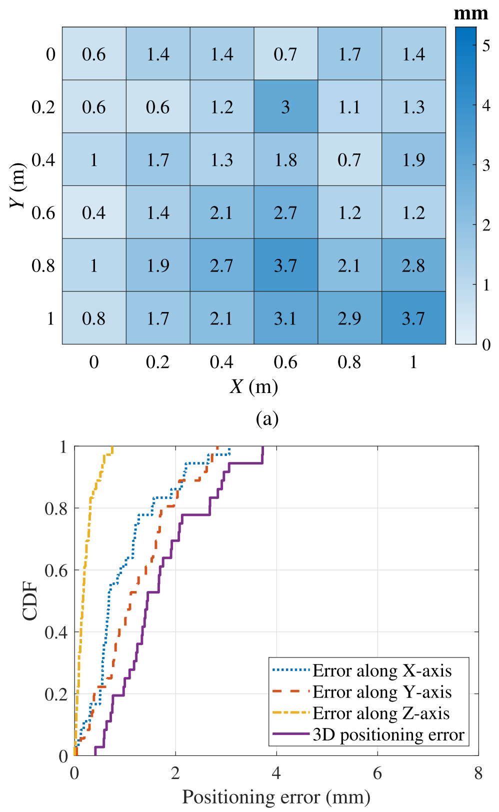


Fig. 14. (a) Distribution of 3D positioning errors. (b) The CDF of positioning errors.

system using 2 cameras and 3 cameras, but 3 cameras achieve significantly lower positioning error along Z-axis than that of 2 cameras.

In addition, we verify the effectiveness of Theorem 4 with experimental parameters, where image noises are modeled as zero-mean Gaussian noises with standard deviations of 1 pixel. We calculate the 3D positioning errors of the 36 test points according to Eq. (37), where the mean positioning error at each test point is obtained by averaging 10^3 independent calculations. The theoretical result of mean 3D positioning error is 1.5 mm, which is very close to the experimental result.

VII. CONCLUSION

In this paper, we have proposed an MC-VLP algorithm which estimates the coordinate of mobile terminal using multiple cameras as the receiver. Specifically, both pinhole camera and fish-eye camera can be employed, which have higher accuracy and wider FOV, respectively. We have conducted a theoretical analysis on the positioning error of the MC-VLP algorithm. The positioning errors along the X-axis, Y-axis, and Z-axis are derived, which reveal the effect of the system parameters on the positioning accuracy. In addition, system parameters are optimized to reduce the positioning error. Simulation results show that the positioning error is significantly reduced within a fixed area via optimizing the camera layout and tilt angle. Experimental results show that the mean 3D positioning accuracy is 1.7 mm with 3 pinhole cameras at height 2.5 m.

APPENDIX

A. Proof of Theorem 1

In Fig. 1(a), let $\mathbf{U}_k$ denote the radial vector from point Q to point O_k, and $\mathbf{V}_k$ denote the tangent vector at point O_k of circle Q. The projections of radial vector $\mathbf{U}_k$ and tangent vector $\mathbf{V}_k$ on the XOY plane are denoted as $\hat{\mathbf{U}}_k$ and $\hat{\mathbf{V}}_k$, respectively. Using the coordinates of points Q and O_k, vectors $\hat{\mathbf{U}}_k$ and $\hat{\mathbf{V}}_k$ in the WCS can be expressed as

$$\begin{cases} \hat{\mathbf{U}}_k = [x_{o_k}^w - x_q^w, y_{o_k}^w - y_q^w, 0]^{\mathrm{T}}, \\ \hat{\mathbf{V}}_k = [y_q^w - y_{o_k}^w, x_{o_k}^w - x_q^w, 0]^{\mathrm{T}}. \end{cases} \tag{41}$$

Denote the plane passing through point O_k, vector $\hat{\mathbf{U}}_k$, and vector $\bar{\mathbf{W}}_k$ as radial plane k. The normal vector to radial plane k can be expressed as the cross product of $\hat{\mathbf{U}}_k$ and $\bar{\mathbf{W}}_k$, i.e.,

$$\begin{aligned} \mathbf{n}_k^r &= \hat{\mathbf{U}}_k \times \bar{\mathbf{W}}_k = [A_k^r, B_k^r, C_k^r]^{\mathrm{T}} \\ &= \left[y_{o_k}^w - y_q^w, x_q^w - x_{o_k}^w, \bar{y}_{p_k}^w (x_{o_k}^w - x_q^w) - \bar{x}_{p_k}^w (y_{o_k}^w - y_q^w)\right]^{\mathrm{T}}. \end{aligned} \tag{42}$$

Using the point O_k and normal vector $\mathbf{n}_k^r$, the point-normal form of the equation of the radial plane k is given by

$$A_k^r(x - x_{o_k}^w) + B_k^r(y - y_{o_k}^w) + C_k^r(z - z_{o_k}^w) = 0. \tag{43}$$

Denote the plane passing through point O_k, vector $\hat{\mathbf{V}}_k$, and vector $\bar{\mathbf{W}}_k$ as tangent plane k. The normal vector to tangent plane k can be expressed as the cross product of $\hat{\mathbf{V}}_k$ and $\bar{\mathbf{W}}_k$, i.e.,

$$\begin{aligned} \mathbf{n}_k^t &= \hat{\mathbf{V}}_k \times \bar{\mathbf{W}}_k = \left[A_k^t, B_k^t, C_k^t\right]^{\mathrm{T}} \\ &= \left[x_{o_k}^w - x_q^w, y_{o_k}^w - y_q^w, \bar{y}_{p_k}^w (y_q^w - y_{o_k}^w) - \bar{x}_{p_k}^w (x_{o_k}^w - x_q^w)\right]^{\mathrm{T}}. \end{aligned} \tag{44}$$

Using the point O_k and normal vector $\mathbf{n}_k^t$, the point-normal form of the equation of the tangent plane k is given by

$$A_k^t(x - x_{o_k}^w) + B_k^t(y - y_{o_k}^w) + C_k^t(z - z_{o_k}^w) = 0. \tag{45}$$

Note that the incident ray is the intersection between radial plane and tangent plane, given by

$$\begin{cases} A_k^r x + B_k^r y + C_k^r z = x_{o_k}^w A_k^r + y_{o_k}^w B_k^r + z_{o_k}^w C_k^r, \\ A_k^t x + B_k^t y + C_k^t z = x_{o_k}^w A_k^t + y_{o_k}^w B_k^t + z_{o_k}^w C_k^t. \end{cases} \tag{46}$$

B. Proof of Theorem 2

For the pinhole camera, the coordinate of image point in the ICS can be obtained by substituting Eq. (25) into Eq. (8):

$$\begin{cases} x^i = f\dfrac{x^c}{z^c} = f \cdot \dfrac{H\sin\theta + x^w\cos\theta}{H\cos\theta - x^w\sin\theta}, \\ y^i = f\dfrac{y^c}{z^c} = f \cdot \dfrac{y^w}{H\cos\theta - x^w\sin\theta}. \end{cases} \tag{47}$$

Based on Eq. (47), we have the following equations:

$$\begin{cases} F(x^i,y^i,x^w,y^w) \triangleq f \cdot \dfrac{H\sin\theta + x^w\cos\theta}{H\cos\theta - x^w\sin\theta} - x^i = 0, \\ G(x^i,y^i,x^w,y^w) \triangleq f \cdot \dfrac{y^w}{H\cos\theta - x^w\sin\theta} - y^i = 0. \end{cases} \tag{48}$$

Furthermore, the partial derivatives of F and G with respect to the four variables x^i, y^i, x^w, and y^w are given by

$$\begin{cases} \dfrac{\partial F}{\partial x^w} = \dfrac{fH}{(H\cos\theta - x^w\sin\theta)^2} = \dfrac{fH}{(z^c)^2}, \\ \dfrac{\partial G}{\partial x^w} = \dfrac{fy^w\sin\theta}{(H\cos\theta - x^w\sin\theta)^2} = \dfrac{fy^c\sin\theta}{(z^c)^2}, \\ \dfrac{\partial G}{\partial y^w} = \dfrac{f}{H\cos\theta - x^w\sin\theta} = \dfrac{f}{z^c}, \\ \dfrac{\partial F}{\partial y^w} = \dfrac{\partial F}{\partial y^i} = \dfrac{\partial G}{\partial x^i} = 0, \\ \dfrac{\partial F}{\partial x^i} = \dfrac{\partial G}{\partial y^i} = -1. \end{cases} \tag{49}$$

Then, the Jacobian determinant is given by

$$J = \frac{\partial(F,G)}{\partial(x^w,y^w)} = \begin{vmatrix} \frac{\partial F}{\partial x^w} & \frac{\partial F}{\partial y^w} \\ \frac{\partial G}{\partial x^w} & \frac{\partial G}{\partial y^w} \end{vmatrix} = \frac{f^2H}{(z^c)^3}. \tag{50}$$

According to the implicit differentiation of equations, we can obtain the following partial derivatives

$$\begin{cases} \dfrac{\partial x^w}{\partial x^i} = -\dfrac{1}{J}\dfrac{\partial(F,G)}{\partial(x^i,y^w)} = -\dfrac{1}{J}\begin{vmatrix} \frac{\partial F}{\partial x^i} & \frac{\partial F}{\partial y^w} \\ \frac{\partial G}{\partial x^i} & \frac{\partial G}{\partial y^w} \end{vmatrix} = \dfrac{(z^c)^2}{fH}, \\ \dfrac{\partial y^w}{\partial x^i} = -\dfrac{1}{J}\dfrac{\partial(F,G)}{\partial(x^w,x^i)} = -\dfrac{1}{J}\begin{vmatrix} \frac{\partial F}{\partial x^w} & \frac{\partial F}{\partial x^i} \\ \frac{\partial G}{\partial x^w} & \frac{\partial G}{\partial x^i} \end{vmatrix} = -\dfrac{y^cz^c\sin\theta}{fH}, \\ \dfrac{\partial x^w}{\partial y^i} = -\dfrac{1}{J}\dfrac{\partial(F,G)}{\partial(y^i,y^w)} = -\dfrac{1}{J}\begin{vmatrix} \frac{\partial F}{\partial y^i} & \frac{\partial F}{\partial y^w} \\ \frac{\partial G}{\partial y^i} & \frac{\partial G}{\partial y^w} \end{vmatrix} = 0, \\ \dfrac{\partial y^w}{\partial y^i} = -\dfrac{1}{J}\dfrac{\partial(F,G)}{\partial(x^w,y^i)} = -\dfrac{1}{J}\begin{vmatrix} \frac{\partial F}{\partial x^w} & \frac{\partial F}{\partial y^i} \\ \frac{\partial G}{\partial x^w} & \frac{\partial G}{\partial y^i} \end{vmatrix} = \dfrac{z^c}{f}. \end{cases} \tag{51}$$

Since the image noises δx^i and δy^i are small amounts, the observation errors δx^w and δy^w are also small amounts. For the pinhole camera, according to Eq. (51), we have

$$\begin{bmatrix} \delta x^w \\ \delta y^w \end{bmatrix} = \begin{bmatrix} \frac{\partial x^w}{\partial x^i} & \frac{\partial x^w}{\partial y^i} \\ \frac{\partial y^w}{\partial x^i} & \frac{\partial y^w}{\partial y^i} \end{bmatrix} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix} = \begin{bmatrix} \frac{(z^c)^2}{fH} & 0 \\ -\frac{y^cz^c\sin\theta}{fH} & \frac{z^c}{f} \end{bmatrix} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix}. \tag{52}$$

C. Proof of Theorem 3

Using the coordinate of the observation point $P = (x^c, y^c, z^c)$ in the CCS, the angle between the incident ray and the Z^c axis can be expressed as

$$\omega = \arctan\frac{\sqrt{(x^c)^2 + (y^c)^2}}{z^c}, \tag{53}$$

where (x^c, y^c, z^c) is given by Eq. (25).

For the fish-eye camera, the distance between the image point and principal point can be expressed as

$$r = \mathcal{F}(\omega) = \mathcal{F}\left[\arctan\frac{\sqrt{(x^c)^2 + (y^c)^2}}{z^c}\right]. \tag{54}$$

The coordinate of image point in the ICS is given by

$$\begin{aligned} x^i &= \frac{rx^c}{\sqrt{(x^c)^2 + (y^c)^2}}, \\ y^i &= \frac{ry^c}{\sqrt{(x^c)^2 + (y^c)^2}}. \end{aligned} \tag{55}$$

Based on Eq. (55), we have the following equations:

$$\begin{cases} F(x^i,y^i,x^w,y^w) \triangleq \dfrac{rx^c}{\sqrt{(x^c)^2 + (y^c)^2}} - x^i = 0, \\ G(x^i,y^i,x^w,y^w) \triangleq \dfrac{ry^c}{\sqrt{(x^c)^2 + (y^c)^2}} - y^i = 0. \end{cases} \tag{56}$$

In order to simplify the subsequent equations, we denote $\xi = (x^c)^2 + (y^c)^2$ and $\eta = x^cz^c\xi^{-1}\cos\theta + \sin\theta$. Then, we have

$$\begin{cases} \dfrac{\partial F}{\partial x^w} = (y^c)^2\xi^{-\frac{3}{2}}\cos\theta\mathcal{F}(\omega) + \dfrac{x^c\eta}{\xi + (z^c)^2}\mathcal{F}'(\omega), \\ \dfrac{\partial F}{\partial y^w} = -x^cy^c\xi^{-\frac{3}{2}}\mathcal{F}(\omega) + \dfrac{x^cy^cz^c\xi^{-1}}{\xi + (z^c)^2}\mathcal{F}'(\omega), \\ \dfrac{\partial G}{\partial x^w} = -x^cy^c\xi^{-\frac{3}{2}}\cos\theta\mathcal{F}(\omega) + \dfrac{y^c\eta}{\xi + (z^c)^2}\mathcal{F}'(\omega), \\ \dfrac{\partial G}{\partial y^w} = (x^c)^2\xi^{-\frac{3}{2}}\mathcal{F}(\omega) + \dfrac{(y^c)^2z^c\xi^{-1}}{\xi + (z^c)^2}\mathcal{F}'(\omega), \\ \dfrac{\partial F}{\partial y^i} = \dfrac{\partial G}{\partial x^i} = 0, \quad \dfrac{\partial F}{\partial x^i} = \dfrac{\partial G}{\partial y^i} = -1. \end{cases} \tag{57}$$

The Jacobian determinant is given by

$$J = \frac{\partial(F,G)}{\partial(x^w,y^w)} = \begin{vmatrix} \frac{\partial F}{\partial x^w} & \frac{\partial F}{\partial y^w} \\ \frac{\partial G}{\partial x^w} & \frac{\partial G}{\partial y^w} \end{vmatrix} = \frac{H\xi^{-\frac{1}{2}}}{\xi + (z^c)^2}\mathcal{F}(\omega)\mathcal{F}'(\omega). \tag{58}$$

Following the same derivation step for the pinhole camera given in appendix B, the relationships between the image noises and the observation errors for the fish-eye camera is given by

$$\begin{bmatrix} \delta x^w \\ \delta y^w \end{bmatrix} = \begin{bmatrix} \frac{\partial x^w}{\partial x^i} & \frac{\partial x^w}{\partial y^i} \\ \frac{\partial y^w}{\partial x^i} & \frac{\partial y^w}{\partial y^i} \end{bmatrix} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix} = \begin{bmatrix} \frac{1}{J}\frac{\partial G}{\partial y^w} & -\frac{1}{J}\frac{\partial F}{\partial y^w} \\ -\frac{1}{J}\frac{\partial G}{\partial x^w} & \frac{1}{J}\frac{\partial F}{\partial x^w} \end{bmatrix} \begin{bmatrix} \delta x^i \\ \delta y^i \end{bmatrix}. \tag{59}$$

Substituting Eq. (57) and Eq. (58) into Eq. (59), we can obtain Eq. (27).

D. Proof of Theorem 4

As shown in Fig. 3(a), we have point $O_k = (R\cos\psi_k, R\sin\psi_k, -H)$ and point $Q = (0,0,-H)$ in the WCS. Using points $\hat{P}_k$ and O_k, the direction vector of the observed incident ray by camera k is given by $\bar{\mathbf{W}}_k = \left[(\hat{s}_k - R\cos\psi_k)/H, (\hat{t}_k - R\sin\psi_k)/H, 1\right]^{\mathrm{T}}$. Then, we can obtain the equation of the observed incident ray according to Theorem 1. To simplify the expression, we multiply the equation coefficients by H/R and obtain

$$\begin{cases} A_k^r = H\sin\psi_k, \\ B_k^r = -H\cos\psi_k, \\ C_k^r = -\hat{s}_k\sin\psi_k + \hat{t}_k\cos\psi_k, \\ E_k^r = H(\hat{s}_k\sin\psi_k - \hat{t}_k\cos\psi_k), \end{cases} \tag{60}$$

$$\begin{cases} A_k^t = H\cos\psi_k, \\ B_k^t = H\sin\psi_k, \\ C_k^t = R - (\hat{s}_k\cos\psi_k + \hat{t}_k\sin\psi_k), \\ E_k^t = H(\hat{s}_k\cos\psi_k + \hat{t}_k\sin\psi_k). \end{cases} \tag{61}$$

According to Eq. (60) and Eq. (61), matrix $\mathbf{F}$ can be expressed as

$$\mathbf{F} = \left[\begin{array}{cc|c} A_1^r & B_1^r & C_1^r \\ \vdots & \vdots & \vdots \\ A_K^r & B_K^r & C_K^r \\ A_1^t & B_1^t & C_1^t \\ \vdots & \vdots & \vdots \\ A_K^t & B_K^t & C_K^t \end{array}\right] = [\mathbf{M}, \mathbf{N}]. \tag{62}$$

To facilitate subsequent calculations, matrix $\mathbf{F}$ is partitioned into submatrices $\mathbf{M}$ and $\mathbf{N}$. In addition, matrix $\mathbf{E}$ can be expressed as

$$\mathbf{E} = [E_1^r, \cdots, E_K^r, E_1^t, \cdots, E_K^t]^{\mathrm{T}}. \tag{63}$$

According to Eq. (29), we define the following four variables:

$$\tau_s = \sum_{k=1}^{K} \hat{s}_k = \sum_{k=1}^{K} (s + \delta s_k) = Ks + \sum_{k=1}^{K} \delta s_k, \tag{64a}$$

$$\tau_t = \sum_{k=1}^{K} \hat{t}_k = \sum_{k=1}^{K} (t + \delta t_k) = Kt + \sum_{k=1}^{K} \delta t_k, \tag{64b}$$

$$\tau_d = \sum_{k=1}^{K} \left[(\hat{s}_k)^2 + (\hat{t}_k)^2\right], \tag{64c}$$

$$\tau_r = \sum_{k=1}^{K} (\hat{s}_k\cos\psi_k + \hat{t}_k\sin\psi_k). \tag{64d}$$

With $\psi_k = \frac{2\pi}{K}(k-1)$ in Eq. (28), we have $\sum\limits_{k=1}^{K} \sin\psi_k = 0$ and $\sum\limits_{k=1}^{K} \cos\psi_k = 0$. In addition, according to Eq. (33), τ_r can be expressed as

$$\tau_r = \sum_{k=1}^{K} (\delta s_k\cos\psi_k + \delta t_k\sin\psi_k) = \sum_{k=1}^{K} \delta g_k. \tag{65}$$

According to Eq. (62) and Eq. (63), each element of the matrices $\mathbf{N}$ and $\mathbf{E}$ contain observation error. Moreover, we have

$$\begin{aligned} \mathbf{F}^{\mathrm{T}}\mathbf{F} &= \begin{bmatrix} \mathbf{M}^{\mathrm{T}} \\ \mathbf{N}^{\mathrm{T}} \end{bmatrix} [\mathbf{M}, \mathbf{N}] = \begin{bmatrix} \mathbf{M}^{\mathrm{T}}\mathbf{M} & \mathbf{M}^{\mathrm{T}}\mathbf{N} \\ \mathbf{N}^{\mathrm{T}}\mathbf{M} & \mathbf{N}^{\mathrm{T}}\mathbf{N} \end{bmatrix} \\ &= \left[\begin{array}{cc|c} KH^2 & 0 & -H\tau_s \\ 0 & KH^2 & -H\tau_t \\ \hline -H\tau_s & -H\tau_t & KR^2 + \tau_d - 2R\tau_r \end{array}\right]. \end{aligned} \tag{66}$$

According to the inverse formula of partitioned matrix, we can obtain

$$\left(\mathbf{F}^{\mathrm{T}}\mathbf{F}\right)^{-1} = \frac{1}{LK^2H^2} \left[\begin{array}{cc|c} KL + \tau_s^2 & \tau_s\tau_t & KH\tau_s \\ \tau_s\tau_t & KL + \tau_t^2 & KH\tau_t \\ \hline KH\tau_s & KH\tau_t & K^2H^2 \end{array}\right], \tag{67}$$

where $L = KR^2 - 2R\tau_r + \tau_d - \frac{1}{K}(\tau_s^2 + \tau_t^2)$.

According to Eq. (64a)-(64c), we define

$$\begin{aligned} \mathcal{M} &= \tau_d - \frac{1}{K}(\tau_s^2 + \tau_t^2) = \\ &\sum_{k=1}^{K} \left[(\delta s_k)^2 + (\delta t_k)^2\right] - \frac{1}{K}\left(\sum_{k=1}^{K} \delta s_k\right)^2 - \frac{1}{K}\left(\sum_{k=1}^{K} \delta t_k\right)^2. \end{aligned} \tag{68}$$

According to Eq. (62) and Eq. (63), we obtain

$$\mathbf{F}^{\mathrm{T}}\mathbf{E} = \begin{bmatrix} \mathbf{M}^{\mathrm{T}} \\ \mathbf{N}^{\mathrm{T}} \end{bmatrix} \mathbf{E} = \begin{bmatrix} \mathbf{M}^{\mathrm{T}}\mathbf{E} \\ \mathbf{N}^{\mathrm{T}}\mathbf{E} \end{bmatrix} = \begin{bmatrix} H^2\tau_s \\ H^2\tau_t \\ -H\tau_d + RH\tau_r \end{bmatrix}. \tag{69}$$

Multiplying Eq. (67) by Eq. (69), the coordinate of the positioning point is estimated as

$$\hat{\mathbf{X}} = \left(\mathbf{F}^{\mathrm{T}}\mathbf{F}\right)^{-1}\mathbf{F}^{\mathrm{T}}\mathbf{E} = \frac{1}{KL} \begin{bmatrix} KR^2\tau_s - R\tau_s\tau_r \\ KR^2\tau_t - R\tau_t\tau_r \\ H(\tau_s^2 + \tau_t^2 - K\tau_d + KR\tau_r) \end{bmatrix}. \tag{70}$$

1) Positioning Error Along X-Axis: Recall that $P = (s, t, 0)$ is the real coordinate of the positioning point. According to Eq. (70), the positioning error along X-axis is given by

$$\Delta X = \frac{KR^2\tau_s - R\tau_s\tau_r}{KL} - s = \frac{R^2\tau_s - R\tau_s\tau_r/K}{KR^2 - 2R\tau_r + \mathcal{M}} - s. \tag{71}$$

As shown in Eq. (68), each term of $\mathcal{M}$ is the square of δs_k or δt_k, so $\mathcal{M}$ is a high-order small quantity. When $\mathcal{M}$ in Eq. (71) is neglected, we have

$$\Delta X \approx \frac{\tau_s(R^2 - R\tau_r/K)}{KR^2 - 2R\tau_r} - s. \tag{72}$$

Substituting Eq. (64a) into Eq. (72), we obtain

$$\Delta X \approx \frac{s\tau_r + (R - \tau_r/K)\sum_{k=1}^{K}\delta s_k}{KR - 2\tau_r}. \tag{73}$$

Noting that $\tau_r \sum\limits_{k=1}^{K} \delta s_k = \sum\limits_{k=1}^{K} \delta g_k \sum\limits_{k=1}^{K} \delta s_k$ is a high-order small quantity, we can neglect this term in Eq. (73) to obtain

$$\Delta X \approx \frac{s\tau_r + R\sum_{k=1}^{K}\delta s_k}{KR - 2\tau_r}. \tag{74}$$

According to Eq. (65), τ_r is the sum of radial component of the observation error, so $KR \gg 2\tau_r$. When $2\tau_r$ in Eq. (74) is neglected, we have

$$\Delta X \approx \frac{s}{KR}\sum_{k=1}^{K}\delta g_k + \frac{1}{K}\sum_{k=1}^{K}\delta s_k. \tag{75}$$

2) Positioning Error Along Y-Axis: According to Eq. (70), the positioning error along Y-axis is given by

$$\Delta Y = \frac{KR^2\tau_t - R\tau_t\tau_r}{KL} - t = \frac{R^2\tau_t - R\tau_t\tau_r/K}{KR^2 - 2R\tau_r + \mathcal{M}} - t. \tag{76}$$

Similar to the derivation of the positioning error along X-axis, Eq. (76) can be simplified as

$$\Delta Y \approx \frac{t}{KR}\sum_{k=1}^{K}\delta g_k + \frac{1}{K}\sum_{k=1}^{K}\delta t_k. \tag{77}$$

3) Positioning Error Along Z-Axis: According to Eq. (70), the positioning error along Z-axis is given by

$$\Delta Z = \frac{H(\tau_s^2 + \tau_t^2 - K\tau_d + KR\tau_r)}{KL} = \frac{H(R\tau_r - \mathcal{M})}{KR^2 - 2R\tau_r + \mathcal{M}}. \tag{78}$$

Neglecting $\mathcal{M}$ in Eq. (78), we have

$$\Delta Z \approx \frac{HR\tau_r}{KR^2 - 2R\tau_r} = \frac{H\tau_r}{KR - 2\tau_r}. \tag{79}$$

Neglecting $2\tau_r$ in Eq. (79), we have

$$\Delta Z \approx \frac{H}{KR}\tau_r = \frac{H}{KR}\sum_{k=1}^{K}\delta g_k. \tag{80}$$

E. Proof of Theorem 5

Define $\tilde{G} = \sum_{k=1}^{K}\delta g_k$, $\tilde{S} = \sum_{k=1}^{K}\delta s_k$, and $\tilde{T} = \sum_{k=1}^{K}\delta t_k$. According to Eq. (37), the expectation of $(\Delta D)^2 = (\Delta X)^2 + (\Delta Y)^2 + (\Delta Z)^2$ can be expressed as

$$E_p = \frac{s^2 + t^2 + H^2}{K^2R^2}\mathbb{E}(\tilde{G}^2) + \frac{1}{K^2}\left[\mathbb{E}(\tilde{S}^2) + \mathbb{E}(\tilde{T}^2)\right] + \frac{2s}{K^2R}\mathbb{E}(\tilde{G}\tilde{S}) + \frac{2t}{K^2R}\mathbb{E}(\tilde{G}\tilde{T}). \tag{81}$$

According to Eq. (32) and Eq. (34), we have $\tilde{G} = \sum_{k=1}^{K}\left(\bar{A}_k\delta x^i + \bar{B}_k\delta y^i\right)$, $\tilde{S} = \sum_{k=1}^{K}\left(\bar{M}_k\delta x^i + \bar{N}_k\delta y^i\right)$, and $\tilde{T} = \sum_{k=1}^{K}\left(\bar{U}_k\delta x^i + \bar{V}_k\delta y^i\right)$. According to Eq. (24), we have $\mathbb{E}(\delta x^i) = 0, \mathbb{E}(\delta y^i) = 0, \mathbb{D}(\delta x^i) = \sigma_x^2$, and $\mathbb{D}(\delta y^i) = \sigma_y^2$, such that $\mathbb{E}(\tilde{G}) = 0$. Since random variables δx^i and δy^i are independent, the expectation of $\tilde{G}^2$ is given by

$$\mathbb{E}(\tilde{G}^2) = \mathbb{D}(\tilde{G}) = \sigma_x^2\sum_{k=1}^{K}\left(\bar{A}_k\right)^2 + \sigma_y^2\sum_{k=1}^{K}\left(\bar{B}_k\right)^2. \tag{82}$$

Similar to the derivation of $\mathbb{E}(\tilde{G}^2)$, we have

$$\mathbb{E}(\tilde{S}^2) = \sigma_x^2\sum_{k=1}^{K}\left(\bar{M}_k\right)^2 + \sigma_y^2\sum_{k=1}^{K}\left(\bar{N}_k\right)^2, \tag{83}$$

$$\mathbb{E}(\tilde{T}^2) = \sigma_x^2\sum_{k=1}^{K}\left(\bar{U}_k\right)^2 + \sigma_y^2\sum_{k=1}^{K}\left(\bar{V}_k\right)^2. \tag{84}$$

According to Eq. (34), the sum of Eq. (83) and Eq. (84) can be simplified as

$$\mathbb{E}(\tilde{S}^2) + \mathbb{E}(\tilde{T}^2) = \sigma_x^2\sum_{k=1}^{K}\left[(\bar{A}_k)^2 + (\bar{C}_k)^2\right] + \sigma_y^2\sum_{k=1}^{K}\left[(\bar{B}_k)^2 + (\bar{D}_k)^2\right]. \tag{85}$$

Since random variables δx^i and δy^i are independent, we have

$$\mathbb{E}(\tilde{G}\tilde{S}) = \mathbb{E}\left[\sum_{k=1}^{K}\left(\bar{A}_k\delta x^i\right)\sum_{k=1}^{K}\left(\bar{M}_k\delta x^i\right)\right] + \mathbb{E}\left[\sum_{k=1}^{K}\left(\bar{B}_k\delta y^i\right)\sum_{k=1}^{K}\left(\bar{N}_k\delta y^i\right)\right]. \tag{86}$$

For different values of k in Eq. (86), random variables $\{\delta x^i\}$ are independent, and random variables $\{\delta y^i\}$ are also independent. According to Eq. (86), we have

$$\begin{aligned}\mathbb{E}(\tilde{G}\tilde{S}) &= \mathbb{E}\left\{\sum_{k=1}^{K}\left[\bar{A}_k\bar{M}_k\left(\delta x^i\right)^2\right]\right\} + \mathbb{E}\left\{\sum_{k=1}^{K}\left[\bar{B}_k\bar{N}_k\left(\delta y^i\right)^2\right]\right\} \\ &= \sigma_x^2\sum_{k=1}^{K}\left(\bar{A}_k\bar{M}_k\right) + \sigma_y^2\sum_{k=1}^{K}\left(\bar{B}_k\bar{N}_k\right).\end{aligned} \tag{87}$$

Similar to the derivation of $\mathbb{E}(\tilde{G}\tilde{S})$, we have

$$\mathbb{E}(\tilde{G}\tilde{T}) = \sigma_x^2\sum_{k=1}^{K}\left(\bar{A}_k\bar{U}_k\right) + \sigma_y^2\sum_{k=1}^{K}\left(\bar{B}_k\bar{V}_k\right). \tag{88}$$

Substituting Eq. (82), Eq. (85), Eq. (87), and Eq. (88) into Eq. (81), E_p can be obtained.

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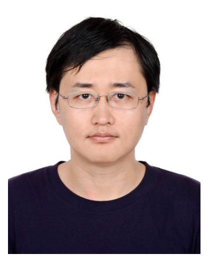

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